Parity Bit Error Detection

Introduction

Parity bit error detection is a simple yet effective method used in digital communication systems to detect errors in transmitted binary data. It is widely used due to its simplicity and low hardware cost.

Objective

To detect whether a transmission error has occurred in binary data using a simple technique involving a single additional bit called the **parity bit**.

Basic Concept

A **parity bit** is an extra bit added to a sequence of binary data bits to ensure that the number of 1s in the string adheres to a predetermined parity rule.

Types of Parity

- **Even Parity**: The number of 1s in the data (including the parity bit) is even.
- **Odd Parity**: The number of 1s in the data (including the parity bit) is odd.

Example

Given the 7-bit data: `1011001`

- **Even Parity**: It already has 4 ones (even), so parity bit = `0`.
  Transmitted: `10110010`
- **Odd Parity**: 4 ones, so to make it odd, parity bit = `1`.
  Transmitted: `10110011`

Analogy: Think of the parity bit like a security guard doing headcount. If everyone is supposed to be in even numbers, and the count is odd, something must be wrong.

How Error Detection Works

1. The sender calculates the parity bit based on the data.
2. The complete data (including the parity bit) is transmitted.
3. The receiver counts the number of 1s:
   - If even parity is used, total number of 1s should be even.
   - If odd parity is used, total number of 1s should be odd.
4. If the rule is violated, an **error is detected**.

Limitation

- Parity can **only detect odd-numbered bit errors**.
- Cannot detect all **multi-bit errors**.
- **Cannot correct** any errors, only detect.

Truth Table for Even Parity (3-bit data)

Data Bits	# of 1s	Parity Bit	Transmitted
000	0	0	0000
001	1	1	0011
010	1	1	0101
011	2	0	0110
100	1	1	1001
101	2	0	1010
110	2	0	1100
111	3	1	1111

Extended Example: Error Detection in Practice

- Transmitted with even parity: `11010011` (parity bit is 1 to make total number of 1s even)
- Received: `11000011` (only 3 ones now)
- **Error detected!** Number of 1s is odd in even parity system.

Visualization: Bit Count Example

Bit Position	1	1	0	1
Count of 1s	3 (Odd)			
Parity Bit	1 (For Even Parity)			

Pros and Cons

Advantages

- Very **easy to implement**.
- Only **1 extra bit** required.
- Good for **low-noise** channels.

Limitations

- **Cannot detect all multi-bit errors**.
- **No correction capability**.

Real-Life Use Cases

- UART communication (microcontrollers)
- Satellite telemetry systems
- Industrial sensor links with low bandwidth

Summary

- Parity bits are a **basic method for detecting errors**.
- Easy to apply for **single-bit error detection**.
- Useful in simple, low-cost systems but **not reliable for robust error handling**.